

TIC 394805902

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 8.130327 d
Epoch (t0) 1601.2674 BTJD
Duration 0.1200 d
Depth 9.252 % (92,516 ppm)
SNR 936.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 7 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 274.8 (gate ≥ 7.0)
Odd/even z-score 0.13 (consistent)
Secondary eclipse detected, phase 0.41, 7.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 11 light-curve dip in this sector: 153.1 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 49.73 (gate ≥ 5.0) Compactness 0.88 (gate ≥ 0.50)
Dimming centroid offset 0.10 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 394805879
0.93 px away, dTmag +1.09 - farther from the dimming than the target

